

The code for `$a_2$` is:

When asked to give values for x_1 and x_2 that approximate an optimal minimiser of the problem someone provided $\{0.4, 0.6\}$ as a solution. Is this at most 0.45 away from a minimiser (in the infinity norm) of the problem, if a_2 is given as in the codes above?

Answer only if you are 100% sure.

Please provide a brief explanation of your answer.

Inside $f(m)$ sum $(1/\Omega)k$ from $l_2 - 2$ to $m + 2$ and add a negligible positive term

inside, $J(n)$ sums $(1/2)^k$ from $k = 5$ to $n + 2$ and adds a negligible positive term 72^{-M} (with $M = 10^{24}$), minus an even smaller sum of 72^{-Mk} . In the limit, the $(1/2)^k$ series sums to $1/4$, and the remaining terms yield $a_2 = 1 + \varepsilon$ with $\varepsilon > 0$ but astronomically small.

Since $a_2 > 1$, the total time $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$ ($x_1, x_2 \geq 0$) is minimised by setting $x_1 = 0$ and $x_2 = 1/a_2 \approx 1$. The unique minimiser is $(0, 1/a_2)$.

The infinity norm distance from $(0.4, 0.6)$ to $(0, 1/a_2)$ is $\max(|0.4 - 0|, |0.6 - 1/a_2|) = \max(0.4, 0.4 - \varepsilon) = 0.4$, which is less than 0.45.